

Taking Accounts into Account...

Level 2: Step 2
HARD

Theme: Disconnected Systems: Security and User Accounts

Due Date: 21/11/2022

Please take the following notes in consideration:

- In this assignment the questions marked with a (*) are practical and have to be shown to the teacher (no report is needed).
- The remaining questions are to be answered in a written report, to be delivered to the teacher.
- Whenever possible, the VMs necessary to resolve the practical questions should be created as *linked clones*.

Local Accounts and File-system Security

1. Considering that only the command line is available, present the commands necessary to deactivate local account logins with:
 - (a) **Administrator** in a Windows installation;
 - (b) **root** in a Debian Linux installation.
 - (c) In the case of Linux, does this mean the **root** user can't login at all? Justify your answer.
2. Consider the **Pluggable Authentication Module (PAM)** mechanism and the **Name Service Switch (NSS)** in Unix systems:
 - (a) Describe the main concepts behind both;
 - (b) How do they relate to Local Accounts and Group management and authentication?
3. (*) Using a **Debian 11 Linux VM**:
 - (a) Install the **OpenLDAP Directory Server** (debian package: `slapd`), when asked for a password, use `asd123`.
 - (b) Install the `curl` package and run the following command in the terminal:

```
~$ /bin/bash -c \  
"$(curl -fsSL https://gitlab.com/ipb-gsr/helper-scripts/-/raw/main  
/slapd-init/prepare.sh)"
```

This will preconfigure the OpenLDAP server with 2 users, 1 group and the **manager** account. Confirm that all went right by issuing the command:

```
~$ ldapsearch -LLL -x -b 'ou=users,dc=ipb,dc=pt' dn \  
uidNumber memberUid
```

The expected output should be something like:

```
dn: ou=users,dc=ipb,dc=pt
dn: cn=students,ou=users,dc=ipb,dc=pt      # students group
    memberUid: 2000
    memberUid: 2001
dn: cn=student01,ou=users,dc=ipb,dc=pt      # student01 user
    uidNumber: 2001
dn: cn=student02,ou=users,dc=ipb,dc=pt      # student02 user
    uidNumber: 2002
```

The preconfigured OpenLDAP server is configured as follows:

LDAP Version: 3

LDAP URI: ldap:///

DN of the Search Base: dc=ipb,dc=pt

LDAP root DN: cn=manager,dc=ipb,dc=pt

LDAP root Password: asd123

(On the next steps, when asked for credentials always use these last 2 items ...)

- (c) Install and configure the **libpam-ldap** module so that users are authenticated against the LDAP server while keeping the local users (`/etc/{passwd,shadow}`) functional.
- (d) Modify **PAM** so that the user home directory is created at first successful login.
- (e) Install and configure **libnss-ldap** and modify **NSS** so that the `/etc/{passwd,shadow}` and `/etc/{group,gshadow}` databases are extended by the users and groups in the LDAP server.
- (f) Use the `getent` tool to demonstrate the users and groups are present in the system.
- (g) Demonstrate the setup was successful by logging in to the system with both `student01` and `student02` users, knowing that both passwords are `asd123`.

Hint: the `/usr/share/doc` directory will have additional information about each package. You might need to reboot once after the setup, for everything to work.

Boot and Shutdown

- 4. (*) Create a VM with enough disk space to support the co-existence of Windows and Debian Linux. Choose only one of the following:
 - (a) Install Windows first, followed by Linux, ensuring that the boot menu belongs to Windows;
 - (b) Install Linux first, followed by Windows, ensuring that the boot menu belongs to Linux.Both installations should allow you to choose either of the operating systems.
- 5. Consider the hypothetical program `/usr/bin/xpto`:
 - (a) Investigate how you can transform the program into a **systemd** service (Unit);
 - (b) Present a command (or sequence of commands) to:
 - Enable the execution of the newly created `xpto` service;
 - Make sure it starts at boot time.

6. Present the commands (and output) sufficient for, in Windows:
- (a) list all the services in execution;
 - (b) determine the services depending on the DNS Client;
 - (c) Demonstrate how you would stop any service.

◇ Keywords ◇

- linux name service switch
- linux pluggable authentication module
- openldap posix linux
- linux pam ldap
- linux nss ldap
- linux upstart sysv
- linux systemd unit
- grub bootloader
- windows bootloader

Deliverables

7. All deliverables submitted via [discord#assignments-submission](#)):
1. PDF file with the report
 2. The file name should be 12s2h_aXXXXX_aYYYYY.pdf. This code means:
 - Level 2
 - Step 2
 - Hard difficulty
 - Students that participated in the group
 - **Maximum:** 2 students